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Physics-Supervised Inversion of Indoor Electromagnetic Materials from Multi-View Images and Channel Measurements



NeurIPS 2026 Conference Submission5499 Authors

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Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

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TL;DR: PRISM inverts indoor electromagnetic material parameters from multi-view images and sparse RF channel measurements via a physics-supervised decoder built on a differentiable Fresnel layer, producing simulation-ready digital twins.

Abstract:

Creating functional Digital Twins, simulatable 3D replicas of the real world, is a central challenge in computer vision. Current methods like NeRF produce visually rich but functionally incomplete twins. The key barrier is the lack of underlying material properties (e.g., permittivity, conductivity). Acquiring this information for every point in a scene via non-contact, non-invasive sensing is a primary goal, but it demands solving an ill-posed physical inversion problem. Standard remote signals, like images and radio frequencies (RF), deeply entangle the unknown geometry, ambient field, and target materials. We introduce PRISM (Physics-supervised Reflection-based Inversion of Surface Materials), a framework for dense, non-invasive physical inversion designed to build functional digital twins. Our key insight is a systematic disentanglement strategy. PRISM leverages image-derived geometry as an anchor, which first enables the resolution of the ambient field. By constraining both geometry and field using only non-invasive data, the original ill-posed problem becomes a physics-supervised learning task. This transformation unlocks our core inversion module: a decoder. Guided by ambient RF signals and a differentiable Fresnel reflection layer paired with a frequency-dispersion power-law parameterization, it learns to explicitly output a continuous, spatially-varying field of the scene's underlying material parameters. We validate our framework on high-fidelity synthetic datasets. Experiments on nine synthetic indoor scenes show that PRISM consistently outperforms direct-regression, per-surface, and RF-neural-field baselines, reducing material parameter MRE by $1.5\times$ -- $3\times$ across all protocol-metric combinations. Beyond appearance-only reconstruction, PRISM yields explicit, simulation-ready material fields that extend digital twins toward physics-aware scene representations.

Primary Area: AI/ML for physical sciences (e.g., physics, chemistry, Earth sciences, astronomy)

Secondary Area: AI and network science (e.g., graph neural networks, communication)

Contribution Type: General: Most submissions will fall into this type.

Submission Number: 5499

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07 Jun 2026, 21:50 Program Chairs, Senior Area Chairs, Area Chairs, Reviewer VXzj

Summary:

PRISM is a method for recovering indoor material properties from multi-view images and RF measurements so a scene can be used as a functional digital twin. It first estimates geometry, then the ambient radio field, then uses a physics-based decoder to infer permittivity and conductivity maps. The main contribution is turning a very hard RF inverse problem into a more structured, supervised one, and it outperforms several baselines on synthetic indoor scenes.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths and Weaknesses: PRISM is a well-motivated and fairly original attempt to turn RF material inversion into a physics-supervised learning problem by combining image-derived geometry, ambient-field reconstruction, and a differentiable Fresnel model. Its main strengths are the clear physical decomposition, strong empirical gains over several baselines, and a useful step toward functional digital twins rather than purely visual reconstructions. The paper is also reasonably significant because it targets a hard and practical inverse problem, and the experiments suggest the method generalizes better than purely data-driven alternatives under the paper’s fixed-geometry setting.

The main weaknesses are that the evaluation is limited to synthetic indoor scenes, the geometry is fixed, and the material model assumes a specular Fresnel-style approximation that may not cover more complex real-world RF behavior. The clarity is generally good, but the method is intricate and would benefit from even more intuition about where each stage helps most and how sensitive the pipeline is to errors in geometry or RF calibration.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good

Questions:

What happens under limited or noisier RF measurements? This would show whether the approach is robust enough for practical use, not just high-quality synthetic data.

Limitations:

The authors do address the main limitations well, especially fixed geometry, the ITU-R dispersion prior, and the specular Fresnel assumption. A useful addition would be a clearer discussion of how these assumptions affect deployment outside the synthetic indoor setting and what minimal extension would be needed for real-world use.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No major issues noted.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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